

58.4%

top-level theorem

```
graph TD; A[top-level theorem] --> B[proof step]; A --> C[proof step]; A --> D[proof step];
```

proof step

proof step

proof step

Mizar 60 (ITP 2023) counts *whole top-level theorems and lemmas*
2,896 holdout problems from 57,897 in
MML 1147; learned premises; 420 CPU s

68.8%

lemma in a development

```
graph TD; A[lemma in a development] --> B[goal]; A --> C[goal]; A --> D[goal];
```

goal

goal

goal

Sledgehammer / AFP (ITP 2022)
counts *proof goals as they arise*
5,000 goals from 50 entries; greedy 16 configurations × 30 CPU s; no reconstruction

Different units, premise selection, and budgets.